

us-east-1.console.aws.amazon.com/sagemaker/home?region=us-east-1#/notebooks-and-git-repos

aws [Search] [Alt+S]

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Amazon SageMaker AI Notebooks and Git Repos

Role manager

▼ Applications and IDEs

- SageMaker Studio
- Canvas
- RStudio
- Notebooks**
- TensorBoard
- Profiler
- Partner AI Apps

▼ Model training & customization

- Training & tuning jobs
- JumpStart model hub
- Nova Forge **New**
- HyperPod clusters

► Training plans

► Deployments & inference

► Model governance

► AWS Marketplace resources

▼ Try the new JupyterLab in SageMaker Studio

Try the new JupyterLab in SageMaker Studio

- Launch notebooks in seconds and start coding instantly
- Use the similar underlying compute and storage as your notebook instances to enable more features at the same cost
- Seamlessly perform comprehensive ML and analytics workflows, all in one notebook
- Leverage GenAI-powered coding assistance from Amazon Q Developer and JupyterAI to accelerate development
- Collaborate with your peers in real-time on the same notebook for seamless ideation

Get Started

► How to access JupyterLab in Studio?

Notebook instances Git repositories

Notebook instances Info

Find notebook instances

< 1 > ⚙

	Name	Instance	Creation time	Status	Actions
☐	MyNotebook	mL.m4.xlarge	7/12/2026, 10:11:55 PM	✓ InService	Open Jupyter Open JupyterLab

CloudShell Feedback

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22:18 12/07/2026

← → ↺ mynotebook-chui.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/hr_attrition_income_analysis.ipynb#3

File Edit View Run Kernel Git Tabs Settings Help

/ en_us /

Name	Modified
column_2C_weka.arff	4s ago
column_2C.dat	4s ago
column_3C_weka.arff	4s ago
column_3C.dat	4s ago
hr_attrition_income_an...	4s ago
linear_learner_mnist-C...	25m ago
PythonCheatSheet-Cop...	21m ago
PythonCheatSheet.ipynb	23m ago
Train_example.ipynb	28s ago
WA_Fn-UseC_-HR-Emp...	next yr.

hr_attrition_income_analysis.i × WA_Fn-UseC_-HR-Employee ×

Open in... conda_python3

Progetto ML: IBM HR Attrition

Questo notebook è stato elaborato da Olga Shopieva (ADABI 2025-2026)

Sommario

1. Intro
2. Importazione di librerie e caricamento dei dati
3. Una occhiata alla struttura dei dati
4. Pulizia delle feature
5. Costruiamo Training e Test set
6. Exploratory Data Analysis - EDA
7. Preprocessing
8. Classificazione - predire Attrition (Si/No)
9. Conclusioni e idee per il miglioramento

1. Intro

Usiamo il dataset IBM HR Analytics Employee Attrition & Performance scaricato da kaggle.com <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset/data>.

- 1470 observations
- 35 features

Ogni riga descrive un dipendente dell'azienda. Le colonne contengono informazioni come età, dipartimento, ruolo, istruzione, esperienza lavorativa, soddisfazione sul

Simple 3 conda_python3 Idle Fully initialized

Mode: Command Ln 1, Col 1 hr_attrition_income_analysis.ipynb 1

Cerca

15:44 12/07/2026

Name	Modified
3_4-machinelearning.ipynb	29 min. ago
column_2C_weka.arff	43 min. ago
column_2C.dat	43 min. ago
column_3C_weka.arff	43 min. ago
column_3C.dat	43 min. ago
ibm-hr-employee-attribution.ipynb	next yr.
linear_learner_mnist.ipynb	49 min. ago
PythonCheatSheet.ipynb	49 min. ago
train.py	1 min. ago
WA_Fn-UseC_-HR-Employee-Attri...	13 min. ago

```
Launcher 3_4-machinelearning.ipynb WA_Fn-UseC_-HR-Employee X train.py ibm-hr-employee-attribution.ipynb
+ + + + +
Code
Open in... conda_python3

X_train_enc = pd.get_dummies(X_train, columns=colonne_categoriche, drop_first=True)
X_test_enc = pd.get_dummies(X_test, columns=colonne_categoriche, drop_first=True)
X_test_enc = X_test_enc.reindex(columns=X_train_enc.columns, fill_value=0)

[4]: train_df = X_train_enc.copy()
train_df.insert(0, "Attrition", y_train.values)

test_df = X_test_enc.copy()
test_df.insert(0, "Attrition", y_test.values)

[5]: import boto3, sagemaker, io, os

sess = sagemaker.Session()
bucket = 'c213913a5405381115843604t1w180050224022-labbucket-sfwgpszgbgknq'
prefix = 'hr-attribution'

s3_resource = boto3.Session().resource('s3')

def upload_s3_csv(filename, folder, dataframe):
    csv_buffer = io.StringIO()
    dataframe.to_csv(csv_buffer, index=False)
    s3_resource.Bucket(bucket).Object(os.path.join(prefix, folder, filename)).put(Body=csv_buffer.getvalue())

upload_s3_csv('train.csv', 'train', train_df)
upload_s3_csv('test.csv', 'test', test_df)

sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml

[11]: from sagemaker.sklearn.estimator import SKLearn

train_channel = sagemaker.inputs.TrainingInput(
```

Il mio bucket S3

la funzione per caricare i dati su S3

us-east-1.console.aws.amazon.com/s3/buckets/c213913a5405381115843604t1w180050224022-labbucket-sfwgpgzgbgknq?prefix=hr-attribution%2Ftrain%2F®ion=us-east-1&tab=obje...

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Account and organization

train/

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	 train.csv	csv	July 12, 2026, 22:55:29 (UTC+02:00)	195.1 KB	Standard

Il train set è caricato su S3

CloudShell Feedback

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22:56 12/07/2026

Name	Modified
3_4-machinelearning.ipynb	29 min. ago
column_2C_weka.arff	43 min. ago
column_2C.dat	43 min. ago
column_3C_weka.arff	43 min. ago
column_3C.dat	43 min. ago
ibm-hr-employee-attribution.ipynb	next yr.
linear_learner_mnist.ipynb	49 min. ago
PythonCheatSheet.ipynb	49 min. ago
train.py	11 min. ago
WA_Fn-UseC_-HR-Employee-Attri...	13 min. ago

```

[*]: from sagemaker.sklearn.estimator import SKLearn

train_channel = sagemaker.inputs.TrainingInput(
    f"s3://{bucket}/{prefix}/train/", content_type='text/csv')
test_channel = sagemaker.inputs.TrainingInput(
    f"s3://{bucket}/{prefix}/test/", content_type='text/csv')

output_path = f"s3://{bucket}/{prefix}/output/"

logreg_estimator = SKLearn(
    entry_point='train.py',
    role=sagemaker.get_execution_role(),
    instance_count=1,
    instance_type='ml.m4.xlarge',
    framework_version='1.2-1',
    output_path=output_path,
    hyperparameters={
        'model-type': 'logreg',
        'max-iter': 1000,
        'threshold': 0.6,
    },
)

logreg_estimator.fit({'train': train_channel, 'test': test_channel}, logs=False)

sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
INFO:sagemaker:Creating training-job with name: sagemaker-scikit-learn-2026-07-12-21-03-44-667
Using provided s3_resource

2026-07-12 21:03:45 Starting - Starting the training job..
2026-07-12 21:04:00 Starting - Preparing the instances for training...
    
```

la funzione per
costruire
estimator(container).

Estimator crea un
Training Job

Amazon SageMaker AI Training jobs

Role manager

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Model training & customization

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- Nova Forge New
- HyperPod clusters

Training plans

- Training plans

Deployments & inference

Model governance

Training jobs

Find training jobs



Advanced search

Actions

Create training job

< 1 > ⚙

	Name	Creation time	Duration	Job status	Job secondary status	Warm pool status	Time left
☐	hr-attribution-rf-2026-07-12-21-26-41-879	7/12/2026, 11:26:42 PM	-	InProgress	Downloading		
☐	hr-attribution-rf-2026-07-12-21-23-49-751	7/12/2026, 11:23:50 PM	2 minutes	Completed	Completed		
☐	hr-attribution-logreg-2026-07-12-21-20-57-894	7/12/2026, 11:20:58 PM	2 minutes	Completed	Completed		
☐	sagemaker-scikit-learn-2026-07-12-21-09-46-136	7/12/2026, 11:09:46 PM	2 minutes	Completed	Completed	-	-
☐	sagemaker-scikit-learn-2026-07-12-21-03-44-667	7/12/2026, 11:03:44 PM	2 minutes	Completed	Completed	-	-
☐	sagemaker-scikit-learn-2026-07-12-21-02-44-188	7/12/2026, 11:02:44 PM	2 minutes	Completed	Completed	-	-
☐	sagemaker-xgboost-2026-07-12-20-31-19-464	7/12/2026, 10:31:19 PM	2 minutes	Completed	Completed	-	-

Training jobs che sono stati eseguiti(per ogni modello -LogisticReg e RandFor

- Role manager
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hr-attrition-logreg-2026-07-12-21-20-57-894

Clone

Create model package

Stop

Create model

Esempio di Training Job

Job settings

Job name

hr-attrition-logreg-2026-07-12-21-20-57-894

Status

✓ Completed

[View history](#)

SageMaker metrics time series

Disabled

IAM role ARN

[arn:aws:iam::180050224022:role/c213913a5405381115843604t1w1-SageMakerExecutionRole-fDsVxVXWpevE](#)

ARN

arn:aws:sagemaker:us-east-1:180050224022:training-job/hr-attrition-logreg-2026-07-12-21-20-57-894

Creation time

Jul 12, 2026 21:20 UTC

Last modified time

Jul 12, 2026 21:23 UTC

Training time (seconds)

130

Billable time (seconds)

130

Managed spot training savings

0%

Tuning job source/parent

-

Algorithm

Algorithm ARN

-

Additional volume size (GB)

30

Maximum wait time for managed spot trainings

-

Volume encryption key

-

Training image

683313688378.dkr.ecr.us-east-1.amazonaws.com/sagemaker-scikit-learn:1.2-1-cpu-py3

Maximum runtime (seconds)

86400

Managed spot training

Disabled

Amazon S3

▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

▼ Files

File systems

▼ Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

▼ Storage management and insights

Storage Lens

Batch Operations

Account and organization

c213913a5405381115843604t1w180050224022-labbucket-sfwgpgzgbgknq [Info](#)

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (2)



Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 > ⚙

☐	Name	Type	Last modified	Size	Storage class
☐	hr-attribution/	Folder	-	-	-
☐	lab3/	Folder	-	-	-

modello si salva in questa
cartella S3

- Amazon S3
- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems
- ▼ Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- ▼ Storage management and insights
 - Storage Lens
 - Batch Operations
- Account and organization

output/

Copy S3 URI

Objects Properties

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	<u>model.tar.gz</u>	gz	July 12, 2026, 23:12:32 (UTC+02:00)	1.5 MB	Standard